HAL-E + AAE Executive Whitepaper

Architecture, Implementation, and Deployment Guide

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Table of Contents

1. Executive Summary

The Hekmati Enterprise Cognitive Cluster (HECC) is a purpose-built cognitive infrastructure system designed to solve a fundamental and persistent problem in professional and academic skill acquisition: the gap between passive knowledge exposure and durable, operational mastery. Most learning environments — whether self-directed, institutional, or corporate — optimize for content delivery, not for the structural construction of expertise. The result is learners who have been exposed to material but who cannot perform reliably under pressure, in novel contexts, or across integrated multi-domain scenarios. HECC addresses this gap directly through two complementary engines — HAL-E (Hyper Accelerated Learning Engine) and AAE (Assessment Automation Engine) — operating as a closed-loop cognitive pipeline in which conceptual construction and diagnostic stress-testing reinforce each other continuously.

HAL-E and AAE are not parallel tools that happen to operate in the same environment. They are architecturally co-dependent. HAL-E builds the conceptual schema — the interconnected mental model of a domain — through structured synthesis, active sparring, invariant anchoring, scenario reasoning, and schema reconstruction. AAE tests that schema under blind diagnostic conditions, tracks mastery score, stability, and drift across sessions, and routes all failure modes back into HAL-E for targeted repair. The result is a closed, bi-directional pipeline in which every failure surfaces a specific repair action, and every repair strengthens the architecture that the next diagnostic session will test. No output is wasted; no session is isolated. Every interaction advances the cartridge — the living knowledge artifact that encodes the learner's domain architecture in portable, open markdown.

Version 1.2 introduces five major additions to the HECC system that collectively bring it from an operational architecture to a fully specified, deployable cognitive infrastructure platform. The Offline-First Cognitive Pipeline formalizes the principle that approximately 80% of all HECC operations — drilling, schema review, error logging, cartridge maintenance — must run offline on local models, with online escalation reserved for the roughly 20% of sessions requiring long-context cross-domain synthesis. The Cockpit Architecture v1.1 specifies the complete operational control layer: device configuration, local model runtime (Ollama), artifact storage structure, and session management protocol. The generalized Model Layer Architecture replaces the prior single-model reference with formalized support for a full local model family (Gemma2, Llama, Mistral, Phi, Qwen) and named online escalation targets (Grok, NotebookLM, Gemini, Claude, Copilot). The Telemetry Schema formalizes the data structure that AAE generates and HAL-E consumes for error routing. The Escalation Protocol introduces a formal decision framework governing transitions from local to online model within a session. Additionally, v1.2 introduces the Partner Licensing Pathways (Tier 1, 2, and 3), nonprofit and consulting firm deployment profiles, and the HAL-E + AAE Family Brand System.

The design philosophy of HECC rests on six governing principles that are non-negotiable invariants of the system itself: schema-first (interconnected understanding over isolated facts), active over passive (teaching back and sparring as primary modalities), depth before breadth (one domain mastered architecturally is operationally worth more than five domains skimmed), hybrid offline-first (local model operations as the default, online escalation as the exception), ownership and portability (all artifacts in open markdown with full learner ownership), and honest telemetry (mastery measured by performance under blind conditions, not self-reported confidence). These principles are not aspirational guidelines — they are structural constraints that shape every component of the system from cartridge file structure to session initialization protocol.

The long-range vision for HECC is a scalable cognitive infrastructure platform that serves the full spectrum from solo learner to institutional deployment. A solo learner uses the complete HECC Core system to achieve durable mastery of a professional certification domain. A cohort program deploys shared cartridge libraries with facilitator-led sparring. A nonprofit workforce development organization runs the full system on free local model infrastructure with no SaaS dependency. A consulting firm deploys firm-authored proprietary cartridges alongside standard certification cartridges to accelerate onboarding and certification pass rates. In every context, the core pipeline is identical — HAL-E builds the architecture, AAE tests it, HAL-E repairs it — and the artifacts remain open, portable, and learner-owned. Version 1.2 establishes the complete foundation for this vision.

2. System Overview

2.1 Purpose

HAL-E and AAE are two complementary engines designed to solve distinct but interdependent problems in cognitive skill acquisition. Together they form a closed-loop cognitive pipeline — a complete hybrid architecture for achieving durable domain mastery. HAL-E addresses the construction problem: how does a learner build a deep, interconnected mental model of a complex domain that will hold under pressure and transfer to novel contexts? AAE addresses the verification problem: how does a system measure the actual state of that mental model under blind conditions, track its stability over time, and generate precise diagnostic data for targeted repair?

Engine	Role
HAL-E — Hyper Accelerated Learning Engine	A schema-first cognitive operating system. Builds interconnected mental models through structured synthesis, active sparring, and invariant anchoring. Produces living knowledge artifacts (cartridges) designed to evolve over months and years.
AAE — Assessment Automation Engine	A high-fidelity diagnostic and drilling engine. Stress-tests conceptual architecture under blind conditions, tracks mastery, stability, and drift, and routes failure modes back into HAL-E for targeted repair.

Core System Insight

"HAL-E builds the architecture. AAE tests it. HAL-E repairs and deepens it. This is a closed, bi-directional cognitive pipeline — not two separate tools operating in isolation."

2.2 Design Principles

Six governing principles form the structural foundation of the HECC system. These are not guidelines — they are operational invariants that constrain every design decision, from cartridge file structure to session management protocol.

- **Schema-First:** Interconnected understanding over isolated facts. Every concept must connect to at least two others within the schema map. Knowledge without architecture is fragile under pressure and fails to transfer to novel contexts.
- **Active Over Passive:** Teaching back, sparring, and scenario reasoning are the primary learning modalities — not passive reading or video consumption. Passive exposure creates familiarity, not mastery. The system is designed to produce performance under blind conditions.
- **Depth Before Breadth:** One domain mastered architecturally is operationally worth more than five domains skimmed. The cartridge system is designed to enforce this — a HAL-E cartridge is not deployed until it is complete and validated, and mastery threshold must be reached before domain advancement.
- **Hybrid Offline-First:** Approximately 80% of operations run offline or on local models. Online escalation (~20%) is reserved for deep synthesis sparring and cross-domain integration sessions. This design priority reflects cost efficiency, operational resilience, and learner autonomy — not a compromise.
- **Ownership and Portability:** All artifacts export as open markdown. No proprietary lock-in. Full learner ownership of all knowledge artifacts. Cartridge files are readable in any markdown-compatible environment and carry no platform dependency.
- **Honest Telemetry:** Mastery is measured by performance under blind conditions — not by self-reported confidence or passive review scores. The telemetry schema captures mastery score, stability, drift, and error classification. A score that cannot be replicated under blind conditions is not mastery.

2.3 v1.2 Upgrade Summary

Version 1.2 introduces significant architectural additions across the model layer, infrastructure specification, telemetry, licensing, and brand systems. The table below summarizes the delta between v1.1 and v1.2 across all major areas.

Area	v1.1	v1.2
Model Layer	Single local model reference (Gemma2)	Generalized: Ollama + Gemma2 / Llama / Mistral / Phi / Qwen
Online Escalation	Generic high-capacity model	Named escalation targets: Grok, NotebookLM, Gemini, Claude, Copilot
Infrastructure	Described conceptually	Cockpit Architecture v1.1 fully specified
Telemetry	Referenced but not formalized	Full telemetry schema defined with required fields and data types
Escalation	Informal	Formal escalation protocol with trigger conditions and decision tree
Licensing	Technical notes only	Full partner licensing pathways documented (Tier 1, 2, 3)
Deployment	Solo + cohort	Added: nonprofit and consulting firm deployment profiles
Brand	Not addressed	HAL-E + AAE Family Brand System introduced

3. System Architecture

3.1 The Cognitive Pipeline

The HAL-E ↔ AAE pipeline operates as a continuous closed loop. Each phase produces structured outputs that feed directly into the next. The loop repeats until the mastery threshold is met, at which point the cartridge state is archived and the domain is advanced. The pipeline is not a linear curriculum — it is a feedback-driven diagnostic and repair cycle. The direction of information flow is bi-directional: AAE diagnostic data is as architecturally important as HAL-E schema construction.

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HAL-E SPRINT (Schema Building)

→ MAP: Build or expand the conceptual schema for the domain

→ SPAR: Active teach-back and pressure-testing of concepts

→ INVARIANTS: Lock non-negotiable rules and relationships

→ SCENARIOS: Multi-step integrated reasoning chains (4–6 steps)

→ SCHEMA REDRAW: Reconstruct the map from memory

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AAE REPS (Blind Diagnostic Testing)

→ ITEM BANK: Domain-weighted blind questions and problems

→ TELEMETRY: Mastery score, stability, drift, time-per-question

→ ERROR ROUTING: Conceptual → Deep Dive | Mechanical → Mini-Drill |
Integrated → Scenario

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FEEDBACK LOOP
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→ AAE telemetry feeds directly back into HAL-E cartridge for targeted repair

→ HAL-E refines schema, adds invariants, builds new scenarios from failure modes

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MASTERY THRESHOLD

→ Target: ≥85% mastery | Low drift | No lens confusion

→ Below threshold: Repeat loop with targeted error routing

→ At threshold: Archive cartridge state → Tag revisit triggers → Advance domain

3.2 HAL-E Architecture

Identity: Schema-first cognitive partner. Constructive orientation. HAL-E does not deliver content — it engineers conceptual architecture. Its primary mode is active sparring, not passive explanation. Every HAL-E session produces a concrete cartridge update: a new cross-link, a strengthened invariant, a new scenario, or a repaired schema node.

Primary Functions:

- **Domain Mapping:** Building and expanding the conceptual schema for a domain — identifying core concepts, their relationships, and their cross-lens implications.
- **Synthesis and Cross-Domain Connection:** Identifying and encoding the relationships between concepts across domain boundaries and between financial and managerial lenses.
- **Invariant Anchoring:** Identifying and locking non-negotiable rules, relationships, and high-yield formulas that must be automatic and retrievable under blind conditions without calculation or reference.
- **Scenario Generation:** Creating multi-step integrated reasoning chains (4–6 steps, cross-lens) that require the learner to move fluidly across domain and lens boundaries within a single scenario.

- **Schema Reconstruction:** Memory-based full domain schema redraw from blank — the highest-fidelity test of internalization, conducted at minimum weekly.
- **Error Repair Routing:** Post-AAE telemetry integration — classifying errors, routing to the appropriate repair mechanism, and updating the cartridge based on diagnosed failure modes.

Scope: Subject-level or Core-level. **Lifespan:** Years. A HAL-E cartridge is a living document that evolves across multiple courses, certification levels, and professional engagements — it does not expire with a single exam or course.

Output Artifacts: Living cartridge files (markdown), schema maps, invariant lists, scenario libraries, retention system logs.

3.3 AAE Architecture

Identity: Domain Cognition Operator. Diagnostic orientation. AAE does not teach — it verifies. Its function is to stress-test the schema that HAL-E has constructed, under blind conditions, and generate precise telemetry that feeds back into HAL-E for repair. AAE produces no cartridge updates directly — all updates are mediated by HAL-E based on AAE diagnostic output.

Primary Functions:

- **Adaptive and Full-Length Practice Assessments:** Item delivery calibrated to exam blueprint domain weights, either adaptive (difficulty adjusting in real-time) or full-length (fixed timed assessment mirroring actual exam conditions).
- **Domain-Weighted Item Bank Delivery:** Item selection weighted according to the Blueprint Map and Weighting Vector, ensuring diagnostic sessions accurately reflect actual assessment structure.
- **Mastery, Stability, and Drift Tracking:** Rolling performance measurement across sessions — not a single score, but a trend line that distinguishes stable mastery from volatile performance spikes.
- **Error Classification and Routing:** Classification of every error as Conceptual, Mechanical, or Integrated, with assignment to the appropriate HAL-E repair mechanism.

- **72-Hour Improvement Planning:** On-demand generation of a targeted repair plan for the subsequent 72 hours, based on current error log and mastery gaps identified in the telemetry record.

Scope: Narrow — course- or skill-specific. **Lifespan:** Exam or interview window duration. AAE cartridges are decommissioned after the target assessment event; the learner's schema (HAL-E) persists.

Output Artifacts: Mastery scores, item bank performance logs, drift data, error routing summaries, telemetry session records.

3.4 The Weighting Vector (AAE Domain Architecture)

The Weighting Vector is the numerical distribution across AAE domain codes that mirrors the target exam blueprint. It governs what proportion of items in any given AAE session are drawn from each domain, ensuring that diagnostic pressure accurately reflects the structure of the actual assessment. The Weighting Vector is authored at the cartridge level and must be validated before any cartridge is deployed.

The following table illustrates the Weighting Vector for the D196 reference implementation (Principles of Financial and Managerial Accounting):

Code	Domain	Weight	Description
D1	Financial Accounting	0.15	External reporting fundamentals, GAAP basics
D2	Financial Statements	0.20	Balance Sheet, Income Statement, Statement of Cash Flows
D3	Budgeting & Cash Flows	0.10	Cash budgeting, forecasting, operating projections
D4	Controlling Costs & Profits	0.10	Cost centers, profit centers, performance evaluation
D5	Managerial Accounting	0.25	Internal decision-making, cost behavior, CVP analysis
D6	Costing Methodologies	0.20	Job Order, Process, Activity-Based Costing
—	**Weighting Vector Sum**	**1.00**	Must always equal exactly 1.00

Validation Rule

"The Weighting Vector must always sum to exactly 1.00. Any imbalance flags a cartridge integrity error and blocks deployment."

4. Cartridge Architecture

4.1 HAL-E Cartridge Standard (Files 00–07)

Every HAL-E cartridge is composed of a standardized set of eight core files (00–07), with one optional file (08). This structure is mandatory and non-negotiable for all cartridges — solo, institutional, and partner. The file structure ensures that any cartridge can be loaded, audited, transferred, and updated by any authorized party without requiring knowledge of how the original cartridge was authored.

File	Name	Purpose
00	Master Boot Sequence	System anchoring, identity, command menu, sprint parameters
01	Cartridge Overview	Scope, lifespan, connection to AAE cartridges, usage rules
02	Conceptual Architecture	The fundamental mental model split, core schema, key interconnections
03	Glossary	Relational, not alphabetical. Terms grouped by conceptual cluster with cross-lens links
04	Schema Maps	Visual and text-based maps showing how concepts interconnect across the domain
05	Invariants	Non-negotiable rules, relationships, and high-yield formulas that must be automatic
06	Scenarios	Multi-step integrated reasoning chains (4–6 steps, cross-lens) for deep synthesis
07	Retention Systems	Spaced retrieval, Feynman practice, error journal, schema redraw cadence
08*	Exam Prep Bridge	Optional. Used when transitioning from HAL-E depth mode to AAE exam-prep mode

4.2 HAL-E Cartridge Lifespan

HAL-E cartridges are designed for longevity. A well-built Accounting Core cartridge is intended to serve from introductory principles through intermediate accounting (D103–D105) and into CPA exam preparation — spanning multiple years and multiple domains without requiring full reconstruction. The cartridge evolves through use: new cross-links are added, invariants are strengthened, and scenarios expand based on real failure modes surfaced by AAE telemetry.

This is a fundamental design distinction from traditional study materials, which are course-specific and discarded after the exam. The HAL-E cartridge is a cumulative knowledge artifact. Its value increases with use, not decreases. A cartridge loaded two years after initial authorship should be richer, more interconnected, and more diagnostically precise than it was when first deployed — because it has been continuously updated by real error data and schema expansion sessions.

4.3 AAE Cartridge Standard (Required Files)

AAE cartridges have a separate required file set optimized for diagnostic function. Unlike HAL-E cartridges, which are living knowledge artifacts designed for years of use, AAE cartridges are diagnostic instruments scoped to a specific exam or assessment target. They are decommissioned after the target event and replaced with a new cartridge for the next assessment target.

File	Purpose
Cartridge Loader	Validates cartridge integrity — confirms all required files are present and Weighting Vector is valid (sums to 1.00)
Blueprint Map	Defines exam domains, weights, and domain-level descriptions
Glossary Terms (CSV)	Domain-tagged term bank — all key terms with definitions and domain codes
Item Bank Templates (CSV)	Structured test items — domain, item type, stem, correct logic, difficulty
Scenario Abstracts	4-step integrated reasoning chains for cross-domain testing

Cartridge Status Protocol

"All cartridges carry a status flag: **READY** / **LOADING** / **ERROR**. A cartridge is only deployed in active sessions when status = READY."

4.4 Item Bank Structure

Every item in the AAE item bank is composed of five required fields: Domain, Item Type, Stem, Correct Logic, and Difficulty. Incomplete items are flagged as errors and excluded from delivery until resolved. The following table illustrates three representative items from the D196 item bank.

Domain	Difficulty	Type	Item Stem	Correct Logic Summary
D2	Medium	Conceptual	Which financial statement reports a company's assets, liabilities, and equity at a specific point in time?	The Balance Sheet. It is a point-in-time snapshot. Assets = Liabilities + Equity must hold.
D5	Hard	Procedural	Fixed costs $120,000. Selling price $80/unit. Variable cost $50/unit. Calculate break-even units and break-even sales dollars.	CM/unit = $30. Break-even units = 4,000. Break-even sales = $320,000.
D1	Medium	Procedural	A company provides $5,000 of consulting services in December but receives payment in January. Record the adjusting entry.	Debit Accounts Receivable $5,000 / Credit Service Revenue $5,000. Revenue recognized when earned under accrual basis.

5. The Daily Crosswalk Ritual

5.1 Overview

The Daily Crosswalk Ritual is the operational core of the HAL-E ↔ AAE pipeline. It is a structured six-step sequence designed to prevent drift, reinforce cross-domain connections, and ensure that conceptual depth and diagnostic performance advance in lockstep. Each step produces a concrete output — a schema update, a sparring gap identified, a scenario completed, a set of telemetry data — that feeds directly into the next step and into the cartridge record. The Crosswalk is not optional on active study days. It is the operational heartbeat of the system.

The Crosswalk is named for the bi-directional movement it enforces between the two engines. A learner who only runs HAL-E sessions develops conceptual architecture that is never stress-tested. A learner who only runs AAE sessions accumulates drilling data without the schema repair that converts error signals into architectural improvement. The Crosswalk enforces both, in sequence, every day.

5.2 The Six Steps

Step 1 — Schema Expansion (HAL-E): Expand the domain schema map by adding at least one new concept or one new cross-link between the financial (external) and managerial (internal) lenses. Schema expansion is the foundation that all subsequent steps test against. Do not proceed to Step 2 until the expansion is logged in File 04 of the active cartridge.

Step 2 — Teach-Back Sparring (HAL-E): Select one concept — ideally one that was added or expanded in Step 1, or one flagged by the previous session's error routing — and explain it from scratch as if teaching a competent but uninformed peer. Note any vagueness, hesitation, or gaps in real time. These are the direct inputs for error routing in Step 5. Vagueness in explanation is a diagnostic signal equivalent to a wrong answer in AAE.

Step 3 — Integrated Scenario (HAL-E): Run one multi-step scenario (4–6 steps) that forces cross-lens reasoning. The scenario must require the learner to move fluidly between financial reporting implications and managerial decision-making implications of the same business event. Scenarios that stay within a single lens do not satisfy Step 3. Cross-lens movement is mandatory.

Step 4 — Blind Micro-Drill (AAE): Run a short, domain-weighted blind testing session (10–20 items). No references, no review material open. The blind condition is non-negotiable. Open-note drilling produces false mastery signals — the learner performs at reference-level competence, not schema-level competence. Only blind-condition scores are diagnostically valid for mastery tracking purposes.

Step 5 — Error Routing (HAL-E): Review AAE telemetry from Step 4. Classify and route each error: Conceptual errors route to schema maps and invariants (File 04 and File 05); Mechanical errors route to a focused mini-drill on the specific formula or procedure; Integrated errors route to an additional cross-lens scenario (File 06). All errors are logged in the error journal (File 07) with the three required fields: invariant violated, lens misread, real-world implication.

Step 6 — Retention Review (HAL-E): Spaced retrieval pass over prior material — specifically material from previous sessions, not today's content. The spaced retrieval schedule is maintained in File 07. Execute a full domain schema redraw from memory at least once per week, outside of the normal Crosswalk sequence. The redraw is the highest-fidelity test of internalization in the system.

5.3 Crosswalk Timing (Recommended Cadence)

Session Component	Duration	Frequency
Schema Expansion	10–15 min	Daily
Teach-Back Sparring	15–20 min	Daily
Integrated Scenario	20–30 min	Daily
AAE Micro-Drill	20–40 min	Daily
Error Routing	10–15 min	Daily
Retention Review	15–20 min	Every 3 days
Full Schema Redraw	30–45 min	Weekly
AAE Full Practice Assessment	90–120 min	End of sprint or as needed

6. Error Routing Architecture

6.1 Error Classification

Error routing is the mechanism by which AAE diagnostic data translates into targeted HAL-E repair actions. Every error produced in an AAE session must be classified into one of three error types before a routing decision is made. Unclassified errors are not routable — they are logged as "unresolved" and escalated to the next Crosswalk session for manual classification.

Error Type	Definition	HAL-E Response
Conceptual	Misunderstood or disconnected mental model — the learner does not correctly understand why a rule applies, what a concept means, or how it connects to adjacent concepts in the schema.	Deep Dive — return to schema maps and invariants (Files 04 and 05). Do not drill until the concept is rebuilt. Drilling a misunderstood concept reinforces the wrong model.
Mechanical	Correct concept, wrong calculation or procedure — the learner understands the principle but cannot execute the formula, procedure, or calculation correctly under pressure.	Mini-Drill — focused AAE item bank session targeting that specific formula or procedure. Concept does not need to be revisited — only the execution mechanism.
Integrated	Failure to connect concepts across lenses or across steps in a multi-step scenario — the learner understands individual concepts but cannot integrate them under cross-domain or multi-step pressure.	Scenario — build or run a multi-step cross-lens scenario in HAL-E (File 06) targeting the integration point that failed. Schema map should also be reviewed for the missing cross-link.

Diagnostic Note

"Misclassifying a Conceptual error as Mechanical is the most common routing failure. If a learner cannot explain *why* a formula applies — not just how to execute it — the error is Conceptual. When in doubt, route to Deep Dive first."

6.2 Error Journal Protocol

Every error logged in the error journal (File 07) must include three required fields. Partial entries are not accepted — an error entry without all three fields is not complete and cannot be used for cartridge updates or telemetry reporting.

- **Invariant Violated:** Which non-negotiable rule or concept was misunderstood or misapplied? This must reference a specific invariant in File 05, not a generic description of the error.
- **Lens Misread:** Which lens was confused — External/Financial vs. Internal/Managerial? If the error did not involve lens confusion, this field is marked "N/A." If lens confusion was present, the specific confusion (e.g., "applied managerial cost classification to external reporting context") must be documented.
- **Real-World Implication:** How would this error manifest in actual financial statements or managerial decisions? This field forces the learner to translate the abstract diagnostic into a concrete operational consequence, which accelerates schema repair.

7. Retention Systems

7.1 Core Retention Mechanisms

The HECC retention architecture is multi-mechanism by design. No single retention technique produces durable mastery across all knowledge types. The six mechanisms below operate in combination, scheduled and managed through File 07 of the active HAL-E cartridge.

Mechanism	Description	Operational Rule
Spaced Retrieval Practice	AAE micro-drills on a rotating schedule targeting high-yield invariants and formulas	If you cannot explain it without notes, it is not retained
Schema Mapping	Maintaining and expanding living schema maps weekly. Monthly full redraw from memory	The map grows with the cartridge and should reflect actual schema complexity
Feynman Technique	Weekly explanation of one major topic as if teaching a beginner. Self-record and note gaps	Vagueness in explanation is the signal, not just wrong answers
Interleaved Practice	Mix problems across sub-domains within a single AAE session	Blocked practice produces weaker transfer than interleaved delivery — interleaving is the default AAE delivery mode
Error Journal & Failure Mode Analysis	Documented errors with invariant violations, lens analysis, and real-world implications	Reviewed before every major assessment — the error journal is a predictive instrument, not a historical record
Future Re-Entry Tagging	Mark high-leverage topics for revisit triggers at later stages of study	Tags stored in File 07 and surfaced when the relevant cartridge is next loaded

7.2 Mastery Threshold

Mastery is not a single score on a single session. It is a sustained state — a configuration of three simultaneous conditions that must all be met and held across multiple sessions before domain advancement is triggered. A single peak score that is not replicated does not constitute mastery; it constitutes a performance spike, which the drift tracking system is specifically designed to flag.

Mastery is declared stable when all three of the following conditions are met simultaneously:

- **≥85% AAE score** across the domain-weighted item bank, sustained across multiple consecutive sessions — not a single peak performance.

- **Low drift** — performance stable across sessions, not spiking and dropping. The rolling 7-session average must remain above the threshold with variance below the drift flag threshold (>10 points deviation).
- **No lens confusion** — financial and managerial implications correctly distinguished without prompting, across both HAL-E sparring sessions and AAE item bank responses.

Status	Action
Below Threshold	Continue the crosswalk ritual with targeted error routing. Do not advance domain.
At Threshold	Archive current cartridge state, tag revisit triggers, export markdown artifacts, advance domain.

8. Offline-First Cognitive Pipeline NEW v1.2

8.1 Design Philosophy

The HECC is architected as an offline-first cognitive system. This is not a fallback mode — it is the primary operating mode. The offline-first design reflects three strategic priorities: cost efficiency, operational resilience, and learner autonomy. The majority of high-frequency, repetitive operations (drilling, schema review, error logging, cartridge maintenance) do not require high-capacity model access and should never depend on it.

The 80/20 split is a design invariant, not a suggestion. Systems that invert this ratio — defaulting to online escalation for routine operations — accumulate unnecessary cost, introduce latency, and create dependency on external infrastructure that disrupts study continuity when unavailable. The offline-first architecture ensures that a learner can run a complete, high-quality Crosswalk Ritual session even without internet access, without a premium model subscription, and without any external infrastructure dependency beyond a local device running Ollama.

8.2 Offline Operation Scope

The following operations are designed to run entirely offline on local models via the Ollama runtime. These represent the approximately 80% of HECC operational volume.

- All AAE drilling sessions — micro-drills and full practice assessments
- Schema review and expansion (Steps 1 and 6 of the Crosswalk Ritual)
- Error logging and error journal maintenance
- Cartridge file maintenance, updates, and integrity checks
- Item bank delivery and domain-weighted question generation
- Retention review and spaced retrieval passes
- Feynman explanation sessions with local model sparring partner

- Cartridge loader validation and status protocol enforcement

8.3 Online Escalation Scope

The following operations require online escalation to a high-capacity model. These represent the approximately 20% of HECC operational volume that cannot be adequately served by the local model layer.

- Deep synthesis sparring requiring long-context reasoning across the full domain schema
- Complex multi-domain scenario generation requiring cross-cartridge integration
- Novel invariant discovery at the intersection of multiple domains
- Integrated error analysis for persistent or anomalous failure patterns that do not resolve through standard routing
- Cross-domain schema reconstruction sessions initiated at cartridge archival or domain transition

8.4 Offline-First Benefits

Cost Control: Local model inference via Ollama is free at the point of use after initial hardware setup. A learner running a complete Crosswalk Ritual in offline mode has zero marginal cost per session. Online escalation is reserved for high-value sessions where the cost is justified by the session objective.

Resilience: An offline-first system cannot be disrupted by API outages, rate limiting, subscription lapses, or internet connectivity failures. The Crosswalk Ritual runs regardless of external infrastructure state. This is particularly important for cohort deployments in educational environments where network access may be inconsistent.

Speed: Local model inference eliminates network round-trip latency for high-frequency operations. Item bank delivery, error routing decisions, and schema review sessions operate at local inference speed, which for 7B–9B models on modern hardware is sufficient for real-time sparring and drilling interactions.

Learner Autonomy: An offline-first architecture reinforces the ownership and portability principle. The learner's cognitive infrastructure — cartridges, item banks, error journals, telemetry logs — runs on their own hardware, under their own control, with no dependency on any external platform's continued availability or pricing structure.

9. Cockpit Architecture v1.1 NEW v1.2

9.1 Overview

The Cockpit Architecture is the operational control layer of the Hekmati Enterprise Cognitive Cluster. It defines the physical and logical environment in which HAL-E and AAE sessions are conducted: the device configuration, model runtime stack, artifact storage layer, and session management protocol. The Cockpit Architecture ensures that every session begins from a consistent, validated state — eliminating setup friction and reducing configuration errors that corrupt telemetry or produce invalid session records.

The term "Cockpit" is intentional. A pilot does not begin a flight without a pre-flight checklist that validates every critical system. A HECC session does not begin without a pre-session checklist that validates the model runtime, cartridge status, error journal state, and session target. Deviation from the initialization sequence is the most common source of session-level data corruption in cognitive systems operating at this level of diagnostic precision.

9.2 Cockpit Components

Layer 1 — Device Configuration

Primary device: Any modern laptop or desktop capable of running local LLMs via Ollama. Minimum recommended specifications: 16GB RAM for 7B–9B parameter models; 32GB RAM for 13B+ parameter models. GPU acceleration (NVIDIA CUDA or Apple Metal) is recommended for 7B–9B models to achieve real-time inference speeds, but is not required — CPU inference is functional for sparring and drilling at these model sizes. SSD storage strongly recommended for model loading performance.

Secondary device (optional): Tablet or phone for flashcard review and error journal access only — not for active sparring or drilling sessions. Qualitative review activities on a secondary device do not count as Crosswalk Ritual steps and do not generate valid telemetry records.

Layer 2 — Local Model Runtime (Ollama)

Ollama serves as the local inference layer for all offline HAL-E and AAE operations. Ollama provides a standardized API interface compatible with all supported local models, enabling model-agnostic cartridge design. Session initialization includes an Ollama service status check as part of the Cartridge Status Protocol — a session cannot proceed to READY status if Ollama is not running or if the designated model is not loaded.

Supported models via Ollama: Gemma2 (9B, 27B), Llama 3.x (8B, 70B), Mistral/Mixtral (7B, 8x7B), Phi-3/Phi-4 (3.8B, 14B), Qwen2.5 (7B, 14B, 72B). Full model specifications and use-case guidance are provided in Section 10.

Layer 3 — Artifact Storage

All HAL-E cartridge files are stored as open markdown in a local directory structure mirrored to a cloud sync provider (e.g., Proton Drive, OneDrive, iCloud Drive). Daily export to the cloud sync target is required — it is not optional and is enforced by the post-session checklist.

Directory structure:

- `/HECC/cartridges/[domain-code]/[file-00 through file-07].md`
- `/HECC/aae/[course-code]/item-bank.csv`
- `/HECC/aae/[course-code]/telemetry/[session-id].json`
- `/HECC/aae/[course-code]/error-log.csv`

Layer 4 — Session Management

Pre-session checklist (enforced before every Crosswalk Ritual):

- Ollama service running and model loaded
- Active cartridge status = READY
- Error journal from previous session reviewed
- Today's domain target confirmed

Post-session tasks (enforced after every session):

- Error journal updated with all new entries from the session
- Telemetry logged to the session record file
- Cartridge files exported to cloud sync target
- Next session domain target noted in cartridge overview (File 01)

9.3 Cockpit Initialization Sequence

- **Power on and environment check:** Confirm device is on primary hardware. Confirm no background processes that will interfere with model inference (e.g., video rendering, large downloads).
- **Launch Ollama:** Start Ollama service. Confirm service is active via `ollama list` or equivalent status check.
- **Load target model:** Pull or confirm the designated local model for today's session is available. For deep sparring sessions, load the highest-parameter model available within hardware constraints.
- **Open cartridge directory:** Navigate to the active cartridge directory for today's domain target. Confirm all 8 required files (00–07) are present.
- **Run Cartridge Loader:** Execute the AAE Cartridge Loader validation check. Confirm status = READY. If status = ERROR, resolve integrity issue before proceeding.
- **Review error journal:** Open File 07. Review all entries from the previous session. Identify priority repair targets for today's Step 5 error routing.
- **Confirm session type and target:** Identify today's Crosswalk step sequence, time allocation, and domain focus. Note any planned online escalation triggers.
- **Open session telemetry record:** Initialize a new session telemetry record with session ID, date, cartridge code, and model tag. This record will be populated throughout the session and finalized in the post-session checklist.

9.4 Cockpit Status Indicators

Indicator	Status	Meaning	Action Required
Ollama Runtime	ACTIVE	Service running, model loaded	None — proceed
Ollama Runtime	INACTIVE	Service not running or model not loaded	Launch Ollama and load model before proceeding
Cartridge Status	READY	All files present, Weighting Vector valid	None — proceed
Cartridge Status	LOADING	Files present but validation in progress	Wait for validation to complete
Cartridge Status	ERROR	Missing files or invalid Weighting Vector	Resolve integrity issue before session launch
Cloud Sync	SYNCED	Last export confirmed by sync provider	None — proceed
Cloud Sync	PENDING	Export queued but not yet confirmed	Confirm sync before closing session
Escalation Target	AVAILABLE	Online model accessible and responsive	None — proceed with escalation
Escalation Target	UNAVAILABLE	Online model unreachable or rate-limited	Defer escalation; continue with local model offline

10. Model Layer Architecture NEW v1.2

10.1 Generalized Local Model Support

v1.2 formalizes support for a generalized local model layer rather than a single reference model. All local models are accessed through the Ollama runtime and are interchangeable at the session level. Cartridge files and protocols are model-agnostic — no cartridge file references a specific model, and switching local models requires no cartridge modification. The only model-specific configuration exists in the Cockpit initialization sequence (model tag selection).

Model Family	Variants Supported	Best Use Cases	Notes
Gemma2 (Google)	9B, 27B	Deep sparring, schema expansion, scenario generation	Strong reasoning architecture; 27B requires 32GB+ RAM. Recommended default for most solo learner deployments.
Llama 3.x (Meta)	8B, 70B	General sparring, teach-back, error routing	70B variant for complex cross-domain synthesis sessions. 8B offers best speed-performance ratio for daily micro-drills.
Mistral / Mixtral	7B, 8x7B	Fast drilling support, item review, error journal assistance	Mixtral 8x7B (MoE architecture) provides richer context windows while maintaining inference efficiency.
Phi-3 / Phi-4 (Microsoft)	3.8B, 14B	Low-resource environments, rapid micro-drill support	Exceptional performance-per-parameter ratio. Recommended for hardware-constrained deployments (8–12GB RAM).
Qwen2.5 (Alibaba)	7B, 14B, 72B	Multilingual deployment, international cohorts	Strong multilingual reasoning. Recommended for nonprofit and international consulting deployments.

Model Selection Guidance

For most solo learners, **Gemma2 9B** or **Llama 3.x 8B** provides the optimal balance of performance and resource efficiency on standard 16GB hardware. The 27B and 70B variants are reserved for deep cross-domain synthesis sessions where long-context reasoning is critical and where inference speed is not the primary constraint.

10.2 Generalized Online Model Escalation

v1.2 defines a named escalation layer for online model access. Each escalation target has a defined use profile within the HECC operational stack. The selection of escalation target is session-objective-driven — the choice should be made at session initialization based on what the session is designed to accomplish.

Escalation Target	Primary Use in HECC	Session Type
Grok (xAI)	Real-time web-grounded synthesis; current-events-adjacent domain updates; regulatory change integration	Deep synthesis sparring with external data grounding
NotebookLM (Google)	Source-grounded deep review; PDF cartridge ingestion and cross-reference; exam guidebook integration	Document-grounded research and cartridge validation sessions
Gemini (Google)	Long-context cross-domain sparring; multimodal schema visualization support; complex scenario generation	Complex scenario generation and cross-domain schema reconstruction
Claude (Anthropic)	Nuanced structured reasoning; long-form scenario elaboration; invariant stress-testing at domain boundary	Deep sparring and scenario design for high-complexity integration challenges
Copilot (Microsoft)	Integrated productivity layer; whitepaper and document generation; spreadsheet-based telemetry tracking and reporting	Artifact management, reporting, and document generation sessions

Escalation Decision Rule

"Escalate online only when the session objective requires capabilities that cannot be delivered by the local model layer: long-context cross-domain synthesis, source-grounded research, or real-time external data integration."

10.3 Model-Agnostic Cartridge Design

All cartridge files are written in plain markdown and are model-agnostic by design. The following design constraints enforce this:

- No cartridge file contains model-specific prompt formats, special tokens, or system prompt syntax that is native to a specific model family.
- Switching local models does not require cartridge modification — only the Ollama model tag in the initialization sequence changes.
- Switching online escalation targets does not require cartridge modification — the escalation target is a session-level decision, not a cartridge-level configuration.
- Cartridge files are readable and operable in any markdown-compatible environment, by any model family, without reformatting.

This constraint is a first-class design priority, not a convenience feature. Model families evolve rapidly. A cartridge authored against a single model reference will degrade in value as models are superseded. A model-agnostic cartridge retains its full value regardless of what changes at the model layer.

11. Telemetry Schema & Escalation Protocol NEW v1.2

11.1 Telemetry Schema

v1.2 formalizes the telemetry data structure that AAE generates and HAL-E consumes for error routing and cartridge updates. Every session produces one Session Telemetry Record and zero or more Error Records. All telemetry is stored as JSON in the artifact storage layer and is never discarded — the full session history is the diagnostic foundation of the mastery tracking system.

Session Telemetry Record — required fields per session:

Field	Type	Description
`session_id`	String (UUID)	Unique identifier for the session
`session_date`	Date (ISO 8601)	Date of session execution
`cartridge_code`	String	Domain code of the active cartridge (e.g., D196)
`model_used`	String	Local model tag or online escalation target name
`session_type`	Enum	MICRO_DRILL / FULL_ASSESSMENT / SPARRING / SCENARIO
`items_delivered`	Integer	Total items delivered in the session
`items_correct`	Integer	Total items answered correctly
`raw_score`	Float (0.00–1.00)	Percentage correct across all items
`domain_scores`	JSON Object	Score broken down by domain code {D1: 0.80, D2: 0.65, ...}
`time_per_item_avg`	Float (seconds)	Average time spent per item
`error_log`	Array	List of Error Records (see schema below) for all items answered incorrectly
`drift_flag`	Boolean	True if session score deviates >10 points from rolling 7-session average
`mastery_status`	Enum	BELOW_THRESHOLD / APPROACHING / AT_THRESHOLD / ARCHIVED

Error Record — required fields per error entry:

Field	Type	Description
`item_id`	String	Reference to the item bank item that produced the error
`domain_code`	String	Domain of the failed item (e.g., D2)
`error_type`	Enum	CONCEPTUAL / MECHANICAL / INTEGRATED
`invariant_violated`	String	Name or description of the invariant misapplied
`lens_misread`	Boolean	True if the error involved financial/managerial lens confusion
`routing_action`	Enum	DEEP_DIVE / MINI_DRILL / SCENARIO

11.2 Escalation Protocol

v1.2 introduces a formal escalation protocol governing the transition from offline (local model) to online (escalated model) within a session. Escalation is a structured decision — not an arbitrary choice made when the local model "feels insufficient." The protocol defines specific trigger conditions, recommended escalation targets, and an exit rule that governs session closure.

Escalation Trigger Conditions — any of the following triggers online escalation:

Trigger	Condition	Escalation Target
Persistent conceptual error	Same invariant appearing in error log across 3+ sessions without resolution	Claude or Gemini (deep sparring)
Cross-domain integration failure	Integrated errors appearing in >2 domain codes in a single session	Gemini or Grok (cross-domain synthesis)
Schema reconstruction failure	Full schema redraw produces <60% of expected nodes from memory	Gemini (long-context schema rebuilding)
Novel domain entry	First session in a completely new cartridge domain with no prior schema	Claude or Gemini (initial schema construction)
Source-grounded review needed	Cartridge update requires verification against current source material or regulatory documentation	NotebookLM (source-grounded review)
Artifact generation	Session produces a document, report, whitepaper, or spreadsheet artifact	Copilot (artifact management)

Escalation Decision Tree:

- Review current session telemetry record. Has a trigger condition been met?
- If YES: Identify the applicable trigger condition and the recommended escalation target from the table above.

- Is the recommended escalation target currently AVAILABLE (confirmed in Cockpit status)?
- If YES: Initiate escalation session. Document the trigger condition, selected target, and session objective in the telemetry record before beginning the escalated session.
- If NO: Select the next-best escalation target from the table. If no target is available, defer the escalation trigger to the next session and continue with the local model for the current session scope.
- Execute the escalated session to its defined objective. The session objective must be specific and achievable within the session — "deep sparring" is not an objective; "rebuild invariant #5 (Matching Principle) and generate two new cross-lens scenarios testing it" is.
- On session close: Confirm that the session produced a concrete cartridge update. If yes, close escalation and return to local model for remaining session components. If no, extend the escalated session to produce the required artifact before closing.

Escalation Exit Rule

"Online escalation sessions must produce a concrete cartridge update — a new invariant, a new scenario, a schema patch, or an error journal entry — before the session closes. Escalation that produces no cartridge artifact is an unproductive use of online capacity and must be documented as an escalation failure in the telemetry record."

11.3 Telemetry Review Cadence

Review Type	Frequency	Action
Session-level review	After every session	Update error journal, finalize and save telemetry record
Weekly drift review	Weekly	Compare rolling 7-session average; flag drift; review error pattern trends
Sprint-end review	End of each sprint	Full domain score analysis; cartridge update decision; escalation frequency audit
Mastery threshold review	When score approaches 85%	Run 3 consecutive sessions to confirm stability; verify no drift flag; confirm no lens confusion

12. Hybrid Workflow Architecture

12.1 Offline vs. Online Split

The HECC hybrid workflow architecture formalizes the operational boundary between local model operations and online escalation across all session types. The approximately 80/20 split is a system-level target, not a per-session target — individual sessions may be 100% offline (a standard micro-drill Crosswalk) or 100% online (a deep synthesis escalation session), but the aggregate operational mix across a sprint should approximate the 80/20 design target.

Mode	Allocation	Primary Use Cases
Offline (local model)	~80%	AAE drilling, schema review, item bank practice, error logging, cartridge maintenance, retention review, Feynman sessions, schema redraw
Online (escalated model)	~20%	HAL-E deep sparring, complex scenario generation, cross-domain synthesis, integrated error analysis, source-grounded review, artifact generation

12.2 Artifact Management

All HAL-E cartridge files are maintained as open markdown throughout their lifespan. The artifact management protocol enforces three operational guarantees that are non-negotiable regardless of deployment context.

- **No Platform Lock-In:** Cartridges are readable and operable in any markdown-compatible environment — text editors, knowledge management systems, LLM chat interfaces, version control repositories, or plain file system directories. No proprietary format, encoding, or container is permitted.
- **Full Learner Ownership:** The learner's knowledge architecture is stored locally and independently. No third-party platform holds the canonical copy of a learner's cartridge. Cloud sync is a backup and portability mechanism, not the primary storage layer.
- **Toolchain Compatibility:** Markdown artifacts are compatible with any future delivery system — web interfaces, mobile apps, document generators, or AI systems not yet in existence. The format choice is a future-proofing decision, not a convenience choice.

13. Domain Coverage: Accounting Core (Reference Implementation)

13.1 The Dual-Lens Architecture

The Accounting Core reference implementation covers three WGU courses as a unified cognitive domain — not three separate courses requiring three separate study approaches. HAL-E's dual-lens architecture makes this possible by treating all three courses as the same information system viewed from different vantage points at different levels of depth.

- **D196: Principles of Financial and Managerial Accounting** — Introductory dual-lens survey. Both lenses introduced simultaneously from the first session. The dual-lens split is established as a first invariant before any specific content is covered.
- **D102: Financial Accounting** — External/financial lens depth. GAAP, accrual basis, financial statement construction, adjusting entries, full reporting cycle.
- **D101: Managerial Accounting** — Internal/managerial lens depth. Cost behavior, CVP analysis, budgeting, variance analysis, costing methodologies.

Foundational Mental Model

"Accounting is not three separate subjects. It is one information system viewed through two output channels — External (Financial Reporting) and Internal (Managerial Decision-Making). All three courses are the same system viewed from different vantage points at different depths."

13.2 Key Invariants in the Accounting Core

The following seven invariants are non-negotiable rules in the Accounting Core domain. They must be automatic — retrievable under blind conditions without calculation, reference, or prompting. They are encoded in File 05 of the Accounting Core HAL-E cartridge and tested directly in every AAE session.

- **The Accounting Equation:** Assets = Liabilities + Equity. This equation must hold after every transaction without exception. Any transaction that violates this equation is either incorrect or incomplete. This is the foundational structural invariant of all financial reporting.
- **Double-Entry Rule:** Debits = Credits on every transaction. Every transaction has at least two entries — a debit to one account and a credit to another. No single-entry bookkeeping is permitted under GAAP. This invariant is the mechanical implementation of the Accounting Equation.
- **Accrual Basis vs. Cash Basis:** GAAP requires accrual basis for external reporting. Revenue is recognized when earned, not when cash is received. Expenses are recognized when incurred, not when cash is paid. The cash basis may be used internally for some purposes but cannot be used for GAAP-compliant financial statements.
- **Product Costs vs. Period Costs:** Product costs (Direct Materials + Direct Labor + Manufacturing Overhead) attach to the product and flow through inventory to COGS. Period costs (Selling and Administrative expenses) are expensed in the period incurred and never attach to the product. This distinction drives both inventory valuation and income statement construction.
- **Relevance and Reliability:** Only future, incremental, avoidable costs drive internal managerial decisions. Sunk costs are irrelevant to future decisions — they have already been incurred and cannot be recovered. This invariant governs all make-or-buy, accept-or-reject, and keep-or-drop decisions in managerial accounting.
- **Relevant Range Constraint:** Cost behavior assumptions — fixed costs remain fixed, variable costs vary proportionally — hold only within the relevant range of activity. Outside the relevant range, fixed costs may step up and variable cost ratios may change. All CVP analysis and budgeting operates within an assumed relevant range.
- **Matching Principle:** Expenses are matched with revenues in the same accounting period. This is the operational justification for accrual basis accounting and the governing rule for all adjusting entries — accruals, deferrals, prepayments, and depreciation.

13.3 High-Yield Formulas (Must Be Automatic)

The following formulas must be retrievable and executable under blind conditions without reference to notes or formula sheets. They are encoded in File 05 of the Accounting Core cartridge and are directly tested in every AAE session.

Formula Name	Expression
Contribution Margin	Sales − Variable Costs
CM per Unit	Selling Price per Unit − Variable Cost per Unit
CM Ratio	CM ÷ Sales
Break-Even Units	Fixed Costs ÷ CM per Unit
Break-Even Sales ($)	Fixed Costs ÷ CM Ratio
Target Profit Sales ($)	(Fixed Costs + Target Profit) ÷ CM Ratio
Gross Margin	Sales − Cost of Goods Sold (COGS)
Operating Income	Gross Margin − Operating Expenses
Price / Rate Variance	(Actual Price − Standard Price) × Actual Quantity
Efficiency Variance	(Actual Quantity − Standard Quantity) × Standard Price

13.4 Cross-Lens Interconnections

The following cross-lens connections are encoded in File 04 (Schema Maps) of the Accounting Core cartridge. Each connection must be explicitly mapped and tested — they cannot be inferred from studying the two lenses in isolation.

Direction	Connection
Financial → Managerial	Accruals and deferrals determine the accurate product vs. period cost split — the timing of cost recognition under accrual basis directly affects how costs are classified in managerial reports.
Financial → Managerial	Inventory valuation method (FIFO / LIFO / Weighted Average) affects both the Balance Sheet value of inventory and the managerial profitability picture simultaneously — it is not a purely financial decision.
Financial → Managerial	Depreciation method affects both reported net income (financial reporting) and internal cost allocation rates (managerial cost accounting) — a change in depreciation method has consequences across both lenses.

Direction	Connection
Managerial → Financial	Budgeting and variance analysis explain deviations between budget and actual figures that appear in external financial statements — the internal planning process leaves a trace in the external report.
Managerial → Financial	Contribution margin thinking improves cash flow forecasting accuracy — understanding which costs are truly variable improves the precision of the Statement of Cash Flows projections.
Managerial → Financial	Cost behavior understanding helps predict future financial position under different volume scenarios — sensitivity analysis in managerial accounting is a direct input to financial statement forecasting.

14. Governance and Cohort Deployment

14.1 Solo Learner vs. Cohort Adaptation

The HECC system is designed first for solo learners but is architecturally compatible with cohort deployment. Adapting to a cohort context requires four governance additions that preserve the core pipeline integrity while accommodating the human facilitation layer.

- **Facilitator Role:** A human facilitator can partially replace AI sparring in Steps 1–3 of the Crosswalk Ritual (Schema Expansion, Teach-Back Sparring, Integrated Scenario). Peer-to-peer Teach-Back is encouraged — it is diagnostically equivalent to AI sparring for identifying gaps in explanation. AAE drilling (Step 4) and Error Routing (Step 5) remain AI-driven and are not replaced by facilitation.
- **Shared Cartridge Libraries:** Cohort-level HAL-E cartridges must be version-controlled using standard version control practices (e.g., Git or equivalent). Individual learner schema maps (File 04) and error journals (File 07) remain learner-owned and are never merged into the shared cartridge library.
- **Cohort Error Aggregation:** Common failure modes appearing across multiple learners in the same session window become shared teaching moments. The facilitator identifies invariants that are failing at the cohort level and triggers a cohort-level cartridge update (File 05 addition or scenario expansion in File 06). Individual error routing remains learner-specific.
- **Adapted Retention Cadences:** The minimum viable cohort cadence is: daily micro-drill + weekly schema review + bi-weekly full schema redraw. The full solo cadence (daily schema expansion + daily teach-back + daily scenario + daily micro-drill + error routing) is the target but may be distributed across cohort sessions rather than completed in a single sitting.

14.2 Cartridge Governance Standards

The following four governance standards are non-optional for all deployments — solo, institutional, and partner. They are preconditions for cartridge deployment, not post-deployment guidelines.

- **Weighting Vector Validation:** All AAE cartridges must pass Weighting Vector validation (sum = 1.00) before deployment. This validation is executed by the Cartridge Loader as part of the initialization sequence and must return PASS before a session proceeds.
- **Cartridge Status Protocol:** READY / LOADING / ERROR status is enforced before every session. A cartridge in ERROR or LOADING status cannot be used in a live session. The session must wait for the status to resolve before proceeding.
- **Refresh Cycles:** Cartridges must be refreshed when: (a) the exam blueprint for the target assessment is updated by the credentialing body; (b) domain knowledge significantly evolves (e.g., a major GAAP standard update); or (c) at minimum annually for any cartridge that remains in active use.
- **Error Journal Aggregation:** Error journal data from individual sessions feeds continuous cartridge improvement. Sprint-end reviews must include an analysis of error journal patterns to identify invariants or scenarios that require cartridge-level updates.

14.3 Licensing Architecture (Technical Notes — v1.1)

HAL-E cartridges are domain-agnostic and can be authored for any professional, academic, or technical domain where the schema-first, invariant-anchored approach is applicable. AAE cartridges require a validated Blueprint Map and a subject-matter-expert-authored item bank that accurately reflects the target assessment structure. Artifact ownership — schema maps, error journals, telemetry records — remains with the learner or cohort at all times. Full partner licensing pathways are detailed in Section 15.

15. Partner Licensing Pathways NEW v1.2

15.1 Overview

v1.2 introduces a formal partner licensing framework enabling institutional deployment of the HAL-E + AAE system beyond the solo learner context. The licensing architecture preserves the core design principles — open markdown artifacts, learner ownership, model-agnosticism — while defining the boundaries of institutional use, customization rights, and cartridge authorship responsibilities. The framework is designed to enable wide adoption across a range of institutional contexts without compromising the architectural integrity of the system or the ownership rights of the learner.

15.2 Licensing Tiers

Tier	Name	Target	Rights	Responsibilities
Tier 1	Solo License	Individual learners	Full system use; personal cartridge authorship; personal artifact ownership	Self-governed; no cartridge distribution to third parties
Tier 2	Institutional License	Academic institutions, bootcamps, cohort programs	Shared cartridge libraries; facilitator-led cohort deployment; branded implementation within guidelines	Blueprint Map validation; cartridge version control; cohort error aggregation reporting; learner artifact ownership policy compliance
Tier 3	Partner License	Consulting firms, nonprofits, training organizations	White-label cartridge authorship; custom domain development; multi-cohort deployment; branded implementation	Subject matter expert cartridge authorship; Weighting Vector validation for all deployed cartridges; artifact ownership policy compliance; cartridge quality assurance

15.3 Cartridge Authorship Rights

Licensed partners at Tier 2 and Tier 3 have defined authorship rights that govern what they may and may not do with cartridge content and system architecture.

Licensed partners may:

- Author domain-specific cartridges in any professional, academic, or technical field using the 00–07 file structure standard.
- Distribute authored cartridges to enrolled learners or cohort members under their institutional or partner license.
- Customize the Weighting Vector for their specific exam blueprint or assessment target, subject to validation requirements.
- Add domain-specific invariants, scenarios, and glossary terms to cartridge files without approval, provided the core file structure is maintained.

Licensed partners may not:

- Modify the core HECC pipeline architecture or system-level cartridge standards (00–07 file structure, telemetry schema, escalation protocol).
- Claim ownership of an enrolled learner's individual schema maps, error journals, retention system logs, or telemetry records.
- Introduce proprietary lock-in into cartridge formats (e.g., converting from open markdown to a proprietary binary format, a closed platform format, or any format that prevents learner access without a platform subscription).
- Distribute cartridges to parties outside their enrolled learner or cohort population without a separate licensing agreement.

15.4 Revenue and Partnership Models

The HECC licensing framework supports multiple partnership and revenue structures depending on the deployment context and organizational structure of the partner.

- **Institutional Subscription:** Annual or semester-based subscription fee for Tier 2 institutional license, scaled by enrolled learner count. Includes access to standard cartridge library, facilitator training, and cartridge governance support.
- **Per-Cohort Licensing:** Fee per cohort deployment for organizations that run periodic, discrete cohorts rather than continuous enrollment. Suitable for bootcamps, intensive certification prep programs, and corporate training events.
- **Consulting Retainer:** Ongoing consulting engagement for Tier 3 partners requiring custom cartridge authorship, domain expert integration, or multi-cohort deployment architecture design.
- **White-Label Deployment:** Full white-label implementation for Tier 3 partners operating under their own brand identity. The partner's brand is the learner-facing identity; HECC architecture operates as the underlying infrastructure layer.

16. Nonprofit & Consulting Firm Deployment NEW v1.2

16.1 Nonprofit Deployment Profile

The HECC system is specifically well-suited for nonprofit educational organizations operating under resource constraints. The offline-first architecture, open markdown artifacts, and free local model runtime (Ollama) eliminate per-user software licensing costs and reduce infrastructure dependency to near zero. A nonprofit running HECC does not require a SaaS subscription, a cloud-based AI platform, or proprietary software — only hardware capable of running a local model and a cloud sync account for artifact backup.

Target organizations: Workforce development nonprofits, community colleges, professional certification prep programs, adult education providers, reentry workforce training programs.

Infrastructure requirements: Minimal. A laptop or desktop per facilitator (for model hosting and cartridge management), shared access for learners to complete AAE drilling sessions, and a cloud sync account (free tier sufficient) for artifact backup. No SaaS subscription required. No per-seat licensing fee for local model inference.

Facilitator model: One trained facilitator per cohort of up to 20 learners. The facilitator manages the shared cartridge library, aggregates error data across the cohort, leads weekly Teach-Back sparring sessions, and coordinates the mastery threshold review cadence. Facilitators do not need to be subject matter experts — they need to understand the HECC pipeline and cartridge governance standards.

Cartridge sourcing: Nonprofits may author their own domain cartridges using the 00–07 file structure standard with guidance from the HECC Cartridge Lab, or may license pre-authored cartridges from Tier 3 partner organizations with subject matter expertise in the target domain.

Deployment Scale	Est. Monthly Cost per Learner	Primary Cost Driver
5 learners	~$0–$10	Hardware amortization + cloud sync (free tier)
20 learners	~$0–$5	Hardware amortization (shared); facilitator time is the primary real cost
100 learners	~$0–$3	Hardware amortization at scale; optional Tier 3 cartridge licensing

16.2 Consulting Firm Deployment Profile

Consulting firms represent a high-value deployment context: professionals needing rapid, deep mastery of a technical domain under time pressure, with high stakes and clear assessment targets (professional certifications, client engagements, regulatory compliance exams). The HECC system's depth-before-breadth philosophy and schema-first architecture are particularly well-suited to this context — consultants do not need survey-level familiarity; they need operational mastery under pressure.

Target use cases: CPA exam preparation, CFA exam preparation, regulatory compliance training, technical onboarding for client-specific domains, methodology framework mastery for new consultants.

Deployment model: Individual consultant license (Tier 1) for self-directed professionals, or firm-wide cohort license (Tier 2/3) for structured onboarding programs and certification prep cohorts. Both models support the full HECC pipeline without modification.

Cartridge strategy: Two-layer cartridge architecture for consulting contexts. Layer 1: Standard professional certification cartridges (e.g., CPA Accounting Core, CFA Financial Analysis) for externally validated domains. Layer 2: Firm-authored proprietary cartridges encoding the firm's methodology frameworks, client-specific compliance requirements, and domain-specific institutional knowledge. Layer 2 cartridges are Tier 3 partner-authored and are classified as confidential firm intellectual property.

Facilitator model: A senior consultant or L&D lead serves as the cartridge steward — responsible for cartridge governance, version control, and cohort error aggregation. Peer-to-peer Teach-Back sparring within project teams is the primary facilitation mechanism: senior consultants spar with junior consultants on domain invariants during engagement onboarding.

Time-to-mastery advantage: Firms deploying HECC for certification exam preparation can expect meaningfully improved first-attempt pass rates compared to traditional self-study methods, due to the diagnostic precision of AAE telemetry, the targeted error routing that prevents inefficient passive review cycles, and the schema-first architecture that builds transferable expertise rather than exam-specific recall. Onboarding speed for new consultants in complex technical domains is similarly accelerated by the invariant-first, scenario-driven approach — new consultants reach operational competence faster when conceptual architecture is built correctly from the first session.

16.3 Deployment Comparison

Dimension	Solo	Institutional / Cohort	Nonprofit	Consulting Firm
Cost Profile	Near-zero (hardware only)	Institutional license fee + hardware	Near-zero (hardware + optional cartridge licensing)	Individual or firm license + optional Tier 3 retainer
Infrastructure	Single device + cloud sync	Shared cartridge library + LMS integration optional	Shared facilitator device + learner access points	Individual devices or firm-managed deployment
Facilitator Required	No	Yes — cohort facilitator	Yes — one per ~20 learners	Yes — L&D lead or senior consultant as steward
Cartridge Authorship	Self-authored (Tier 1)	Institutional-authored or licensed (Tier 2)	Self-authored or Tier 3 licensed	Standard + firm-authored proprietary (Tier 3)
Artifact Ownership	100% learner	Shared cartridge library (institution) + learner schema maps	Shared cartridge library (nonprofit) + learner schema maps	Standard artifacts: learner; Proprietary Layer 2: firm
Time-to-Mastery Target	Self-paced; threshold-governed	Cohort cadence; facilitator-governed	Cohort cadence; facilitator-governed	Accelerated; engagement-deadline-governed

17. HAL-E + AAE Family Brand System NEW v1.2

17.1 Brand Overview

v1.2 introduces the HAL-E + AAE Family Brand System — a unified identity framework for the Hekmati Enterprise Cognitive Cluster and all derivative products, licensed deployments, and partner implementations. The brand system defines naming conventions, visual identity principles, product family structure, and usage guidelines for institutional and partner contexts. The brand system is designed to be coherent across the full spectrum from solo learner product to white-label partner deployment — the same architectural identity is expressed at every level of the product family.

17.2 Core Brand Identity

System Name: Hekmati Enterprise Cognitive Cluster (HECC)

Engine Names: HAL-E (Hyper Accelerated Learning Engine) | AAE (Assessment Automation Engine)

Brand Voice: Precise, architectural, high-performance. The brand communicates mastery infrastructure, not passive learning tools. Language emphasizes building, testing, routing, and repairing — engineering metaphors for cognitive work. The brand does not use the language of traditional educational technology: no "courses," "content delivery," "learning journeys," or "engagement." The vocabulary is operational: pipelines, cartridges, invariants, telemetry, routing, thresholds.

Brand Promise: "Build the architecture. Test it under pressure. Repair with precision. Achieve durable mastery."

17.3 Product Family Structure

Product Name	Description	Target
HECC Core	Full HAL-E + AAE system with Cockpit Architecture — the complete solo learner deployment with all pipeline components and local model runtime	Individual learners
HECC Cohort	Facilitated cohort deployment with shared cartridge libraries, aggregated telemetry, and facilitator-led sparring integration	Academic institutions, bootcamps, cohort programs

Product Name	Description	Target
HECC Pro	Consulting firm deployment with firm-authored proprietary cartridges, multi-cohort management, and L&D lead stewardship model	Consulting firms, L&D departments
HECC Nonprofit	Reduced-cost deployment optimized for community organizations and workforce development — offline-first, zero SaaS dependency, shared facilitator model	Nonprofits, community colleges, workforce development organizations
HECC Cartridge Lab	A domain cartridge authoring environment — tools, templates, and governance guidelines for Tier 2/3 partners authoring custom domain cartridges	Institutional and partner licensees (Tier 2/3)

17.4 Naming Conventions

All products, licensed implementations, and derivative deployments must comply with the following naming conventions:

- **Core system:** Always referred to as "HECC" (Hekmati Enterprise Cognitive Cluster) in full, or "HECC" in abbreviated form. Never abbreviated differently or combined with other acronyms.
- **Engines:** Always "HAL-E" (with hyphen) and "AAE" — never "HALE," "HAL E," or "Assessment Engine." The hyphen in HAL-E is a required element of the engine name.
- **Licensed implementations:** Partner-branded implementations carry the partner name as a prefix: "[Partner Name] × HECC" for co-branded deployments, or "[Partner Name] Cognitive Cluster powered by HECC" for white-label deployments where the partner brand is primary.
- **Cartridges:** Named by domain code and version: "[Domain Code] Cartridge v[version]" (e.g., "D196 Cartridge v1.2"). Partner-authored cartridges may carry a partner prefix: "[Partner] D196 Cartridge v1.0."
- **Product variants:** HECC product family names (Core, Cohort, Pro, Nonprofit, Cartridge Lab) are always capitalized as proper names when referring to the product.

17.5 Visual Identity Principles

The HECC visual identity is designed to communicate precision instrumentation and architectural depth — not approachable educational software. The following principles govern all visual expression of the brand across documents, interfaces, and partner materials.

- **Color Palette Philosophy:** High-contrast and technical. Primary palette emphasizes deep navy, slate, and off-white with accent colors suggesting precision instrumentation — electric blue for active state indicators, amber for warning/drift flags, green for threshold-achieved states. The aesthetic reference is cockpit instrumentation and systems architecture diagrams, not educational platform UI design.
- **Typography:** Monospace or technical sans-serif (e.g., Consolas) for all system-level labels, code references, command syntax, and telemetry field names. Clean serif or professional sans-serif (Calibri, Cambria) for body text in documents and cartridge files. No decorative or display fonts.
- **Iconography:** Architectural diagrams, pipeline flow charts, circuit-inspired schema maps, and precision indicator graphics. No stock educational imagery (graduation caps, books, lightbulbs, pencils). The visual language is engineering, not classroom.
- **Document Standards:** All HECC documents carry the following metadata in the header: version number, classification level, author, system identifier (HECC), and issue date. Documents without this metadata are not considered official HECC artifacts regardless of content.

17.6 Brand Usage Guidelines for Partners

Licensed partners may:

- Use the "powered by HECC" designation in co-branded or white-label materials, subject to the naming convention requirements in Section 17.4.
- Reference HAL-E and AAE by name in materials delivered to enrolled learners within their licensed deployment.
- Adapt HECC document templates to carry their organizational branding, provided the HECC system identifier and version information remain present.

Licensed partners may not:

- Use the HECC name, HAL-E name, or AAE name in public-facing marketing materials without explicit written approval from the system author.
- Modify the brand voice or brand promise in ways that contradict the core design principles (schema-first, active over passive, offline-first, learner ownership).

- Sub-license the HECC brand to third parties outside their licensed deployment scope without a separate licensing agreement.
- Represent HECC as a passive content delivery platform, a traditional LMS, or any tool that conflicts with the operational identity of the system.

18. Appendix

A. HAL-E Command Reference

Command	Function
`MAP [Concept]`	Build or expand schema map for a concept or domain area. Adds new nodes and cross-links to File 04 of the active cartridge.
`SPAR [Topic]`	Deep sparring session — teach-back and active pressure-testing of understanding. Produces gap identification for error routing.
`INVARIANTS`	Review and reinforce non-negotiable rules for the current domain. Runs through all File 05 entries with active recall testing.
`SCENARIO [Topic]`	Run a multi-step integrated reasoning chain (4–6 steps, cross-lens) on the specified topic. Produces scenario entry for File 06.
`RETENTION REVIEW`	Spaced retrieval pass over prior material from the error journal and schema map. Executes the File 07 spaced retrieval schedule.
`RUN AAE [Subtopic]`	Escalate to AAE for targeted drilling on a specific subtopic or formula cluster. Initiates a domain-weighted micro-drill session.
`LOAD CARTRIDGE`	Load and activate a specific HAL-E cartridge by name or domain code. Executes Cartridge Loader validation.
`SCHEMA REDRAW`	Reconstruct the full domain schema map from memory without reference material. Highest-fidelity internalization test in the system.
`ERROR ROUTE [Error]`	Classify a specified error and route to the appropriate repair mechanism (DEEP_DIVE / MINI_DRILL / SCENARIO). Logs entry to File 07.

B. AAE Command Reference

Command	Function
`Run AAE [Course] INTERACTIVE`	Adaptive session — item difficulty adjusts dynamically based on real-time performance within the session. Produces mastery score and error log.
`Run AAE [Course] FULL`	Full-length practice assessment mirroring actual OA conditions — timed, blind, domain-weighted. Produces complete telemetry record.
`Reinitialize [Course]`	Reload cartridge and reset session state. Use when cartridge has been updated between sessions or when status = ERROR requires resolution.
`RUN TELEMETRY REPORT`	Generate session telemetry summary and drift analysis. Outputs session_id, raw_score, domain_scores, drift_flag, and mastery_status.
`RUN DRIFT CHECK`	Compare current session score against rolling 7-session average. Flags if deviation exceeds 10 percentage points in either direction.
`EXPORT CARTRIDGE`	Export all cartridge files to markdown for cloud sync. Confirms all 8 required files are present and telemetry records are saved.

C. Glossary of System Terms

Term	Definition
HAL-E	Hyper Accelerated Learning Engine. The schema-building, sparring, and synthesis engine within the HECC. Primary function: construct and repair conceptual architecture.
AAE	Assessment Automation Engine. The blind diagnostic testing, drilling, and telemetry engine paired with HAL-E. Primary function: verify schema integrity under blind conditions and generate routing data.
Cartridge	A structured set of markdown files (HAL-E: Files 00–07; AAE: 5 required files) encoding a domain's schema, invariants, item bank, and telemetry configuration. The canonical knowledge artifact of the HECC system.
Weighting Vector	A numerical distribution across AAE domain codes mirroring the exam blueprint. Must sum to exactly 1.00. Validated by the Cartridge Loader before every session.
Invariant	A non-negotiable rule or relationship within a domain that must be automatic — retrievable under pressure without calculation or reference. Encoded in File 05 of the HAL-E cartridge.
Schema Map	A living text- or visual-based map showing how concepts interconnect within and across domain lenses. Encoded in File 04. Grows with use; redrawn from memory at least weekly.
Error Routing	The classification of AAE errors into Conceptual, Mechanical, or Integrated types and assignment to the appropriate HAL-E repair mechanism (DEEP_DIVE / MINI_DRILL / SCENARIO).
Drift	Instability in AAE performance across sessions — spiking and dropping rather than maintaining a stable score. Flagged by the drift_flag field in the telemetry record when session score deviates >10 points from rolling average.
Lens Confusion	Failure to correctly distinguish between financial (external) and managerial (internal) implications of the same business event or accounting concept.
Mastery Threshold	≥85% AAE score sustained across multiple sessions, with low drift and no lens confusion. All three conditions must be met simultaneously before domain advancement.
Crosswalk Ritual	The six-step daily operational sequence advancing HAL-E depth and AAE performance in parallel: Schema Expansion → Teach-Back → Scenario → Micro-Drill → Error Routing → Retention Review.
Feynman Technique	Explanation of a concept from scratch as if teaching a beginner. Gaps and vagueness in explanation feed the error journal as Conceptual error signals.
Interleaved Practice	Mixing problems across sub-domains within a single AAE session to strengthen transfer. The default AAE item delivery mode — blocked practice is not used.
Cockpit Architecture	The operational control layer defining device configuration, model runtime (Ollama), artifact storage structure, and session management protocol for all HECC sessions.
Ollama	The local LLM runtime serving as the offline model layer for all HECC offline operations. Provides standardized API access to all supported local model families.

Term	Definition
Telemetry Schema	The formalized data structure (v1.2) capturing session performance, error records, drift flags, and mastery status per session. Stored as JSON in the artifact storage layer.
Escalation Protocol	The formal decision framework (v1.2) governing transitions from local model to online model within a session, including trigger conditions, target selection, and exit requirements.
HECC	Hekmati Enterprise Cognitive Cluster. The complete system comprising HAL-E, AAE, Cockpit Architecture, and supporting cartridge infrastructure.
Tier 1/2/3 License	The three-tier partner licensing framework: Solo (Tier 1), Institutional (Tier 2), Partner/White-Label (Tier 3). Each tier defines deployment rights, authorship rights, and governance responsibilities.
HECC Cartridge Lab	The cartridge authoring environment and toolset for Tier 2/3 licensed partners. Provides templates, governance guidelines, and validation tools for custom domain cartridge development.

D. Telemetry Schema Reference (v1.2)

The following tables reproduce the full telemetry schema from Section 11 for quick reference during session management and cartridge governance activities.

Session Telemetry Record:

Field	Type	Description
`session_id`	String (UUID)	Unique identifier for the session
`session_date`	Date (ISO 8601)	Date of session execution
`cartridge_code`	String	Domain code of the active cartridge (e.g., D196)
`model_used`	String	Local model tag (e.g., gemma2:9b) or online escalation target name
`session_type`	Enum	MICRO_DRILL / FULL_ASSESSMENT / SPARRING / SCENARIO
`items_delivered`	Integer	Total items delivered in the session
`items_correct`	Integer	Total items answered correctly
`raw_score`	Float (0.00–1.00)	Percentage correct across all items
`domain_scores`	JSON Object	Score broken down by domain code {D1: 0.80, D2: 0.65, ...}
`time_per_item_avg`	Float (seconds)	Average time spent per item
`error_log`	Array	List of Error Records for all items answered incorrectly
`drift_flag`	Boolean	True if session score deviates >10 points from rolling 7-session average
`mastery_status`	Enum	BELOW_THRESHOLD / APPROACHING / AT_THRESHOLD / ARCHIVED

Error Record:

Field	Type	Description
`item_id`	String	Reference to the item bank item that produced the error
`domain_code`	String	Domain of the failed item (e.g., D2)
`error_type`	Enum	CONCEPTUAL / MECHANICAL / INTEGRATED
`invariant_violated`	String	Name or description of the invariant misapplied
`lens_misread`	Boolean	True if the error involved financial/managerial lens confusion

Field	Type	Description
`routing_action`	Enum	DEEP_DIVE / MINI_DRILL / SCENARIO

E. Version History

Version	Date	Author	Summary of Changes
1.0	June 2026	John Hekmati	Initial release — core HAL-E + AAE architecture, accounting domain reference implementation, cartridge standard (Files 00–07), crosswalk ritual, error routing, and retention systems.
1.1	July 2026	John Hekmati	Added cartridge governance standards, cohort deployment guidelines, licensing technical notes (v1.1). Refined weighting vector validation protocol. Expanded domain coverage documentation for Accounting Core.
1.2	July 2026	John Hekmati	Added: Cockpit Architecture v1.1 (Section 9), generalized model layer architecture with full local model family support and named online escalation targets (Section 10), formalized telemetry schema and escalation protocol (Section 11), offline-first cognitive pipeline specification (Section 8), partner licensing pathways — Tier 1/2/3 (Section 15), nonprofit and consulting firm deployment profiles (Section 16), HAL-E + AAE Family Brand System (Section 17). Updated appendix with v1.2 glossary terms and telemetry schema reference.

Hekmati Enterprise Cognitive Cluster — HAL-E + AAE Executive Whitepaper, Version 1.2

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All cartridge artifacts are stored as open markdown. No proprietary lock-in. Artifact ownership remains with the learner.